

# Lilyan Leblanc

Nationality: French

 lilyan.leblanc@gmail.com | Full, clean driving license

## Robotics engineer specialized in surgical applications

**Objective:** To contribute to innovation in surgical robotics to provide state-of-the-art solutions to healthcare professionals

## Professional Experience

### 2021-2025 | PHD STUDENT IN SURGICAL ROBOTICS | INSTITUT DES SYSTEMES INTELLIGENTS ET DE ROBOTIQUE - SORBONNE UNIVERSITE - FRANCE

- Developed and tested a multimodal signal processing method for breach detection in autonomous robotic pedicle screw drilling for spine surgery
- Designed a non-rigid registration method and explored computer vision techniques for spine localization in autonomous robotic pedicle screw drilling
- Contributed to software development using C++, Matlab, ROS2 and Git
- Conducted ex-vivo, cadaveric and in-vivo experiments and documented related results
- Engaged in inter-laboratory collaboration as part of an international research project
- Taught master's students in experimental robotics, control and C programming

### 2018-2021 | MEDICAL ROBOTICS ENGINEER | ZIMMER BIOMET ROBOTICS - FRANCE

- Studied, validated, improved and investigated fabrication process of the ROSA robot
- Solved robotics challenges and automated the calibration process of the ROSA robot in an Agile team using C++ and Git. Contributed to both technical development and team management. Led to a 15% time reduction of the manufacturing process.
- Provided a state-of-the-art of serial robots calibration methods
- Worked in a regulatory environment and contributed to risk analysis, Computer System Validation, NC and CAPA processes (ISO13486, ISO 9001)
- Contributed to lean manufacturing and automation of incoming inspection, production and calibration processes.

## Education & Certifications

### 2021-2025 | PHD IN SURGICAL ROBOTICS | INSTITUT DES SYSTEMES INTELLIGENTS ET DE ROBOTIQUE - SORBONNE UNIVERSITE - FRANCE

### 2023 | UNIVERSITY DEGREE IN BIOLOGICAL AND MEDICAL ENGINEERING | SORBONNE UNIVERSITE - FRANCE

- Funding, protection and translation of biomedical research into practical applications
- Entrepreneurship, funding et innovation strategy in a regulated biomedical environment

### 2016-2018 | MASTER OF SCIENCE IN IMAGING, ROBOTICS AND BIOMEDICAL ENGINEERING | TÉLECOM PHYSIQUE STRASBOURG - STRASBOURG UNIVERSITY - FRANCE

- Valedictorian
- Subjects included: Computer science, Robotics, Automatics, Computer Vision, Image processing, Signal processing

### 2015-2018 | HEALTHCARE INFORMATION TECHNOLOGY ENGINEER | TÉLECOM PHYSIQUE STRASBOURG - STRASBOURG UNIVERSITY - FRANCE

- Valedictorian
- Subjects included: Computer vision, Automatics, Robotics, Computer science, Biomechanics, Electronics, Biology

# Skills

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## LANGUAGE SKILLS

- **English:** Very proficient (C1) – TOEIC: 970/995
- **German:** Independent user (B1-B2)
- **Spanish:** Basic user (A2-B1)

## COMPUTER SKILLS

- **Programming languages:** C, C++, Python
- **Version control:** Git
- **Other (scientific) languages:** MATLAB
- **Framework:** ROS/ROS2
- **Operating systems:** Windows XP to 10, Ubuntu
- **Database:** MySQL, MicrosoftAccess, PostgreSQL

# Research & Innovation

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## PATENTS

- L. Leblanc et al., “*Statistical Methods and Systems For Detecting Perforations During Surgical Drilling Based On A Sensed Electrical Characteristics*”
  - Patent approval pending
  - Spinal canal perforation detection using electrical conductivity signal and Bayesian trend change detection methods

## PUBLICATIONS

- L. Leblanc et al., “*Pedicle Drilling Planning Transfer for Spine Surgery Using Functional Map Correspondences*”. In *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 6947-6952). IEEE.
  - Non-rigid registration method for transferring drilling trajectories in pedicle screw placement
  - Evaluation through numerical simulation
- L. Leblanc et al., “*Automatic Spinal Canal Breach Detection During Pedicle Screw Placement*”, IEEE Robotics and Automation Letters, Vol. 9, Issue 2, pp. 1915-1922, February 2024. DOI: 10.1109/LRA.2024.3349947
  - Spinal canal perforation detection using electrical conductivity signal and Bayesian trend change detection methods
  - Validation with ex-vivo vertebrae drillings

# Activities and interests

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- Advanced **scuba diver** and **master freediver**
- Competitive **triathlon**
- **Acrobatic Rock and Roll** and **contemporary dance**
- **Cinema** (once to twice a week)